

Xiaoyin Chen

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EDUCATION

Mila, Université de Montreal <i>Doctor of Philosophy, Computer Science, Advisor: Prof. Yoshua Bengio</i>	Sept. 2023 – Present GPA: 4.00/4.00
Duke University <i>Master of Science, Computer Science, Advisor: Prof. Sam Wiseman</i>	Aug. 2021 – May 2023 GPA: 4.00/4.00
University of California, Irvine <i>Bachelor of Science, Computer Science, Minor in Statistics</i>	Sept. 2016 – Mar. 2020 GPA: 3.94/4.00, Major GPA: 3.99/4.00

PUBLICATIONS & PREPRINTS

Peer-reviewed conference, journal, and workshop

Learning to Self-Evolve

Xiaoyin Chen, Canwen Xu, Yite Wang, Boyi Liu, Zhewei Yao, Yoshua Bengio, Yuxiong He.
ICLR 2026 Workshop on Lifelong Agents: Learning, Aligning, Evolving, <https://arxiv.org/abs/2603.18620>, 2026.

How Much Backtracking is Enough? Exploring the Interplay of SFT and RL in Enhancing LLM Reasoning

Hongyi James Cai, Junlin Wang, **Xiaoyin Chen**, Bhuwan Dhingra.
Conference on Language Modeling (COLM) 2026, openreview.net/forum?id=bpVWLDwlu4, 2026.

Self-Evolving Curriculum for LLM Reasoning

Xiaoyin Chen, Jiarui Lu, Minsu Kim, Dinghuai Zhang, Jian Tang, Alexandre Piché, Nicolas Gontier, Yoshua Bengio, Ehsan Kamalloo.
Conference on Language Modeling (COLM) 2026, <https://arxiv.org/abs/2505.14970>, 2025.

Bayesian Symbolic Regression with Entropic Reinforcement Learning

Oussama Boussif, Mohammed Mahfoud, Younesse Kaddar, Moksh Jain, Sida Li, Damiano Fornasiere, **Xiaoyin Chen**, Yoshua Bengio, Esmeralda S. Whitamner.
Uncertainty in Artificial Intelligence (UAI) 2026, <https://arxiv.org/abs/2608.09617>, 2026.

When Greedy Wins: Emergent Exploitation Bias in Meta-Bandit LLM Training

Sanxing Chen, **Xiaoyin Chen**, Yukun Huang, Roy Xie, Bhuwan Dhingra.
International Conference on Learning Representations (ICLR) 2026, <https://arxiv.org/abs/2509.24923>, 2025.

Search-Based Correction of Reasoning Chains for Language Models

Minsu Kim, Jean-Pierre Falet, Oliver E. Richardson, **Xiaoyin Chen**, Moksh Jain, Sungjin Ahn, Sungsoo Ahn, Yoshua Bengio.
International Conference on Learning Representations (ICLR) 2026, <https://arxiv.org/abs/2505.11824>, 2025.

PipelineRL: Faster On-policy Reinforcement Learning for Long Sequence Generation

Alexandre Piché, Ehsan Kamalloo, Rafael Pardinas, **Xiaoyin Chen**, Dzmitry Bahdanau.
Transactions on Machine Learning Research (TMLR), <https://arxiv.org/abs/2509.19128>, 2025.

Aligning Protein Conformation Ensemble Generation with Physical Feedback

Jiarui Lu*, **Xiaoyin Chen***, Stephen Zhewen Lu, Aurelie Lozano, Aurelie Lozano, Vijil Chenthamarakshan, Payel Das, Jian Tang.
Forty-Second International Conference on Machine Learning, <https://arxiv.org/abs/2505.24203>, 2025.

Structure Language Models for Protein Conformation Generation

Jiarui Lu*, **Xiaoyin Chen***, Stephen Zhewen Lu, Chence Shi, Hongyu Guo, Yoshua Bengio, Jian Tang.
International Conference on Learning Representations 2025, <https://arxiv.org/abs/2410.18403>, 2024.

HarmAug: Effective Data Augmentation for Knowledge Distillation of Safety Guard Models

Seanie Lee, Haebin Seong, Dong Bok Lee, Minki Kang, **Xiaoyin Chen**, Dominik Wagner, Yoshua Bengio, Juho Lee, Sung Ju Hwang.

International Conference on Learning Representations 2025, <https://arxiv.org/abs/2410.01524>, 2024.

Proof Flow: Preliminary Study on Generative Flow Network Language Model Tuning for Formal Reasoning

Matthew Ho, Vincent Zhu, **Xiaoyin Chen**, Moksh Jain, Nikolay Malkin, Edwin Zhang.

System-2 Reasoning at Scale Workshop at NeurIPS 2024, <https://arxiv.org/abs/2410.13224>, 2024.

Inference and Verbalization Functions During In-Context Learning

Junyi Tao*, **Xiaoyin Chen***, Nelson F. Liu.

Findings of Empirical Methods in Natural Language Processing (EMNLP) 2024, <https://arxiv.org/abs/2410.09349>, 2024.

Efficient Causal Graph Discovery Using Large Language Models

Thomas Jiralerspong*, **Xiaoyin Chen***, Yash More, Vedant Shah, Yoshua Bengio.

AGI Workshop at ICLR 2024, <https://arxiv.org/abs/2402.01207>, 2024.

Learning Consistent Deep Generative Models from Sparse Data via Prediction Constraints

Gabriel Hope, Madina Abdrakhmanova, **Xiaoyin Chen**, Michael C. Hughes, Erik B. Sudderth.

4th Symposium on Advances in Approximate Bayesian Inference, 2022. arxiv.org/abs/2012.06718.

On the Current Failure – But Bright Future – of Topology-driven Biological Network Alignment

Siyue Wang, **Xiaoyin Chen**, Brent J. Frederisy, Benedict A. Mbakogu, Amy D. Kanne, Pasha Khosravi, Wayne B. Hayes.

Advances in Protein Chemistry and Structural Biology: Protein interaction networks, Volume 131.

<https://arxiv.org/abs/2204.11999>.

Cross-species Prediction of Protein Function by Global Network Alignment

Siyue Wang, **Xiaoyin Chen**, Brent J. Frederisy, Benedict A. Mbakogu, Amy D. Kanne, Pasha Khosravi, Giles R.S. Atkinson, Wayne B. Hayes.

28th Conference on Intelligent Systems for Molecular Biology (protein prediction track), 2020.

https://www.iscb.org/cms_addon/conferences/ismb2020/tracks/functioncosi.

Preprints & Under Review

BM25 Query Augmentation Learned End-to-End

Xiaoyin Chen, Sam Wiseman.

<https://arxiv.org/abs/2305.14087>, 2023.

WORK EXPERIENCE

Student Researcher

Google DeepMind, Mountain View, USA

Jun. 2026 – Present

Research Intern

Snowflake Research, Menlo Park, USA

Sep. 2025 – Jun. 2026

Visiting Researcher

ServiceNow Research, Montreal, Canada

Nov. 2024 – Jul. 2025

TEACHING EXPERIENCE

Teaching Assistant

Duke University

- **CS590.05 Spring 2023**: Computational Biology.

Undergrad Tutor & Grader

University of California, Irvine

- **ICS-33 Spring 2017**: Intermediate Programming in Python;
- **ICS-46 Fall 2018**: Data Structure Implementation and Analysis in C++.

SERVICE

Reviewer: ACL, EMNLP, Neurips, ICLR, ICML